

Hospital Management System

Database Design

1. Purpose

A Hospital Management System is designed to manage and organize the daily operations of a hospital. It helps store information about patients, doctors, appointments, admissions, treatments, prescriptions, and payments.

The database ensures that hospital information is stored in a structured manner and can be accessed efficiently when required.

2. Required Tables

The Hospital Management System requires the following tables:

- Patients
- Doctors
- Departments
- Appointments
- Admissions
- Medical_Records
- Prescriptions
- Medicines
- Bills

3. Tables and Their Attributes

Table 1: Patients

Column Name	Data Type	Description
patient_id	INT (PK)	Unique ID of the patient
patient_name	VARCHAR	Name of the patient
date_of_birth	DATE	Date of birth
gender	VARCHAR	Gender of the patient
phone_number	VARCHAR	Contact number
address	VARCHAR	Residential address

Primary Key: patient_id

This table stores personal and contact details of all patients.

Table 2: Departments

Column Name	Data Type	Description
department_id	INT (PK)	Unique department ID
department_name	VARCHAR	Name of the department
location	VARCHAR	Location or floor of department

Primary Key: department_id

This table manages hospital departments such as Cardiology, Neurology, Orthopedics, and General Medicine.

Table 3: Doctors

Column Name	Data Type	Description
doctor_id	INT (PK)	Unique ID of the doctor
doctor_name	VARCHAR	Name of the doctor
department_id	INT (FK)	Department where doctor works
specialization	VARCHAR	Doctor's specialization
phone_number	VARCHAR	Contact number
email	VARCHAR	Email address

Primary Key: doctor_id

Foreign Key: department_id

This table stores professional information about doctors and connects them to their respective departments.

Table 4: Appointments

Column Name	Data Type	Description
appointment_id	INT (PK)	Unique appointment ID
patient_id	INT (FK)	Patient booking the appointment
doctor_id	INT (FK)	Doctor assigned
appointment_date	DATETIME	Date and time of appointment
appointment_status	VARCHAR	Scheduled, Completed, Cancelled

Primary Key: appointment_id

Foreign Keys: patient_id, doctor_id

This table manages patient appointments and consultation schedules.

Table 5: Admissions

Column Name	Data Type	Description
admission_id	INT (PK)	Unique admission ID
patient_id	INT (FK)	Admitted patient
doctor_id	INT (FK)	Doctor responsible
admission_date	DATE	Date of admission
discharge_date	DATE	Date of discharge
room_number	VARCHAR	Assigned room number

Primary Key: admission_id

Foreign Keys: patient_id, doctor_id

This table manages inpatient records, including admission and discharge details.

Table 6: Medical_Records

Column Name	Data Type	Description
record_id	INT (PK)	Unique medical record ID
patient_id	INT (FK)	Related patient
doctor_id	INT (FK)	Doctor handling the case
diagnosis	VARCHAR	Disease or medical condition
treatment	VARCHAR	Treatment provided
record_date	DATE	Date of medical record

Primary Key: record_id

Foreign Keys: patient_id, doctor_id

This table maintains the patient's medical history, diagnosis, and treatment information.

Table 7: Prescriptions

Column Name	Data Type	Description
prescription_id	INT (PK)	Unique prescription ID
patient_id	INT (FK)	Patient receiving prescription
doctor_id	INT (FK)	Doctor prescribing medicine
prescription_date	DATE	Date of prescription

Primary Key: prescription_id

Foreign Keys: patient_id, doctor_id

This table stores the general details of prescriptions created by doctors.

Table 8: Medicines

Column Name	Data Type	Description
medicine_id	INT (PK)	Unique medicine ID
medicine_name	VARCHAR	Name of medicine
manufacturer	VARCHAR	Medicine manufacturer
price	DECIMAL	Price of medicine
stock_quantity	INT	Available stock

Primary Key: medicine_id

This table manages the information and availability of medicines.

Table 9: Bills

Column Name	Data Type	Description
bill_id	INT (PK)	Unique bill ID
patient_id	INT (FK)	Patient being billed
admission_id	INT (FK)	Related admission, if applicable
total_amount	DECIMAL	Total hospital charges
payment_status	VARCHAR	Paid, Pending, Partially Paid
bill_date	DATE	Date of bill generation

Primary Key: bill_id

Foreign Keys: patient_id, admission_id

This table manages hospital charges and payment information.

4. Relationships Between Tables

Table 1	Relationship	Table 2
Departments	One-to-Many	Doctors
Patients	One-to-Many	Appointments
Doctors	One-to-Many	Appointments
Patients	One-to-Many	Admissions
Doctors	One-to-Many	Admissions
Patients	One-to-Many	Medical_Records
Doctors	One-to-Many	Medical_Records
Patients	One-to-Many	Prescriptions
Doctors	One-to-Many	Prescriptions
Patients	One-to-Many	Bills
Admissions	One-to-One / One-to-Many	Bills

5. Simple Relationship Diagram

DEPARTMENTS → DOCTORS → APPOINTMENTS

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PATIENTS → ADMISSIONS → MEDICAL RECORDS → PRESCRIPTIONS → MEDICINES

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-----BILLS

6. Overall Working of the Hospital Management System

1. A patient registers with the hospital.
2. The patient books an appointment with a doctor.
3. The doctor belongs to a specific department.
4. During consultation, the doctor creates medical records and prescriptions.
5. If necessary, the patient is admitted to the hospital.
6. Admission details and room information are recorded.
7. Medicines prescribed to the patient are managed through the medicine database.
8. Finally, the hospital generates a bill for the services provided.

7. Summary of Tables

Table Name	Main Purpose
Patients	Stores patient information
Departments	Stores hospital department details
Doctors	Stores doctor information
Appointments	Manages doctor-patient appointments
Admissions	Manages patient admissions and discharge
Medical_Records	Stores diagnosis and treatment history
Prescriptions	Stores prescription information
Medicines	Stores medicine details and stock
Bills	Manages hospital billing and payments

Conclusion

The Hospital Management System database efficiently organizes important hospital operations by connecting patients, doctors, departments, appointments, medical records, admissions, prescriptions, medicines, and billing information. Primary keys uniquely identify each record, while foreign keys establish relationships between tables. This makes the system organized, reduces data redundancy, and allows efficient management of hospital activities.